Simulation Arena

Deal Approval & Margin Balancing

No discount makes money on these numbers: even the least-bad one (0%) loses $2,308.54M. The deal is unprofitable before any discount is given - the fix is the cost to deliver or the price, not the size of the concession.

Seed 42 | Run run_dba653f0bc68 | 170,000 scenarios tested | Monte Carlo simulation with an empirica… | 21 Sep 2026, 21:29

Same seed, same numbers - this run can be reproduced exactly.

How this works

For each discount level we ran thousands of 'what-if' worlds. In each one the buyer's odds of saying yes wobble a little, and your delivery cost wobbles a little. We then ask the only question that matters: if you offered this discount again and again, how much profit would you bank on average? That is the expected profit - the chance of winning multiplied by the profit you keep. Because those worlds are a sample, we also work out how much the answer itself would wobble if we ran it again. Any discount whose expected profit is within twice that wobble of the best - or within 1% of the best profit, whichever is wider - is treated as giving the same answer. That is the range where you have genuine room to negotiate.

The numbers that matter

BEST DISCOUNT TO OFFER
0%
No discount makes money here - 0% loses the least.
vs rep asked for 25%

EXPECTED PROFIT AT THAT DISCOUNT
-$2,308.54M
This is a loss, not a profit: on these numbers the deal costs more to deliver than it sells for.

DISCOUNT RANGE THAT LOSES THE LEAST
0%
Only one discount stands out - the model cannot tell any neighbouring discount apart from it. The model's own margin of error here is about $126.13M, so treat this as the best single number - the next discount along already costs real money.
vs rep asked for 25% - outside the range

WHERE IT STARTS TO GENUINELY COST YOU
2%
At 2% the expected profit falls $247.03M below the best. That gap is wider than the model's own margin of error, so the loss is real.
vs costs about $247.03M

EXPECTED PROFIT IF YOU CAVE IN
-$9,777.32M
What 25% actually earns you.
vs -$7,468.79M vs the optimum

PROFIT GIVEN AWAY
$7,468.79M
Buying +49 points of win probability costs you this much.

ODDS OF WINNING AT THE BEST DISCOUNT
36.9%
The chance the deal lands.

WORST CASE (1-IN-20)
-$8,004.55M
In an unlucky run of 20 deals, this is roughly the profit you would see.

How we got there

1. The two forces fighting each other

Every point of discount does two things: it makes the buyer a bit more likely to say yes, and it takes $209.12M straight off your profit. Early on the first effect wins. Later, the second one does.

Evidence: Cost to deliver $27,110.89M on $20,911.98M of list value

2. What the buyer's behaviour looks like

They start around 35.0% likely to buy at full price and never become certain - the curve flattens near 95.0%. Around 17% discount is where they are most persuadable.

Evidence: Curve source: 20000 of your past deals

3. The sweet spot

0% discount loses the least: $2,308.54M on average, with a 36.9% chance of winning. There is no discount here that makes money - the deal is already unprofitable at full price.

Evidence: peak of 17 levels tested

4. The range where the answer is the same

0% is the only discount that stands out; the model cannot tell any neighbouring discount apart from it. The next level along already costs $247.03M.

Evidence: margin of error $126.13M; safe range 0%

5. What the requested discount really costs

Moving from 0% to 25% buys +49 points of win probability but costs $7,468.79M of expected profit - more than the model's margin of error, so the loss is real.

Evidence: -$9,777.32M vs -$2,308.54M

Charts

Where profit peaks

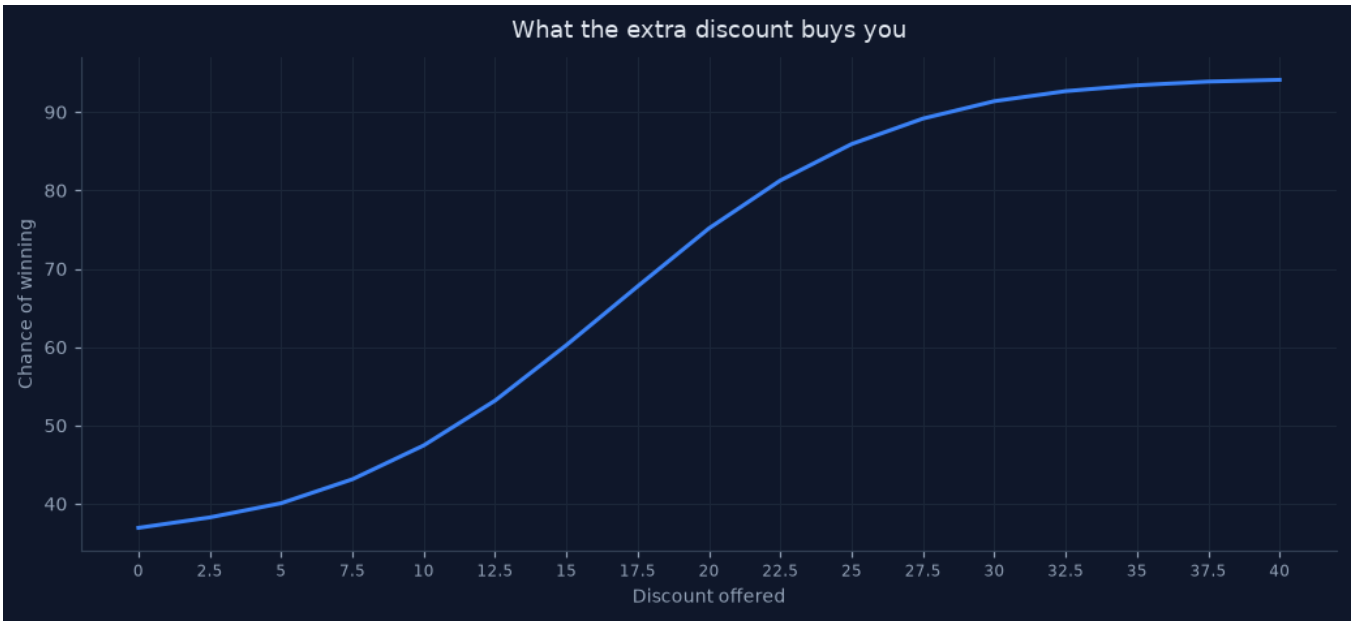
Everything from 0% gives the same answer within the model's margin of error; the peak is 0%



What this shows: Grey is what you keep per won deal - it only falls. Green is what you actually expect to earn - it rises, peaks, then falls. The amber dashed line is the best profit minus the model's own margin of error: every discount whose green line stays above it is the same answer, which is the 0% range. Past that you are genuinely paying for the win.

What the extra discount buys you

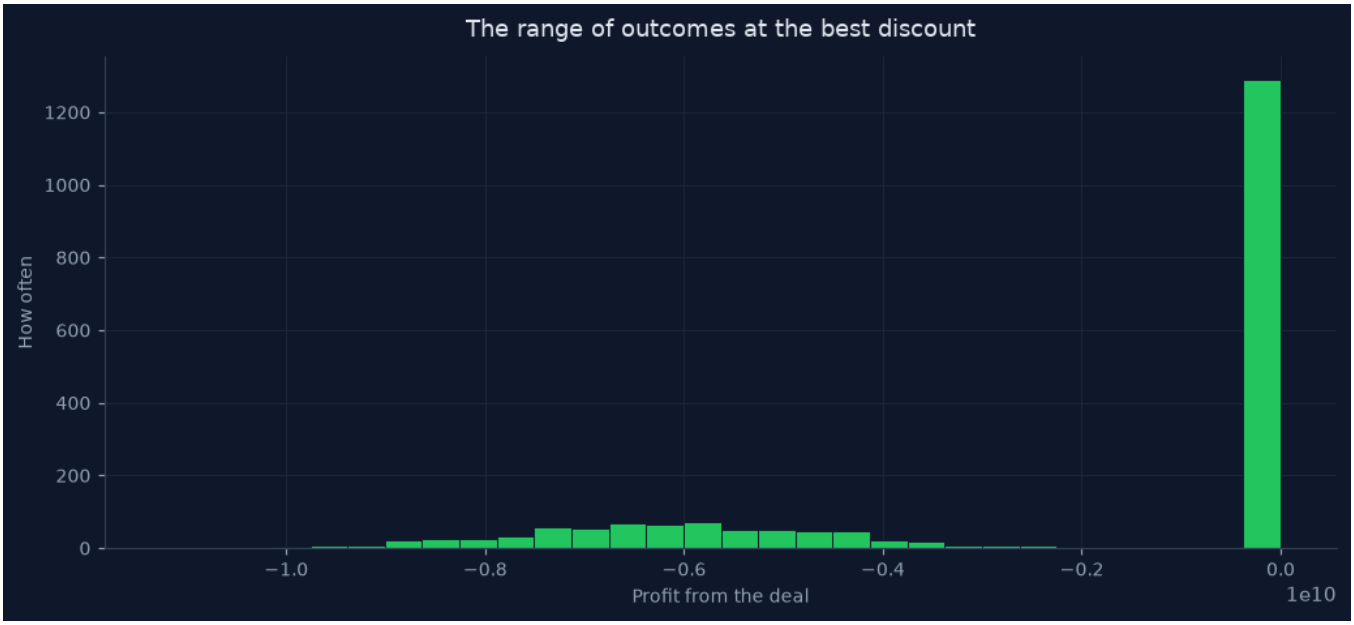
Curve learned from 20000 of your past deals



What this shows: Past the knee of this curve, extra discount buys almost no extra certainty - but keeps costing margin. That is the trap.

The range of outcomes at the best discount

10,000 simulated worlds at 0% discount



What this shows: The cluster at zero is the deals you lose. The spread on the right is the profit you make when you win.

Tables

Every discount level tested

Expected profit = chance of winning x profit kept. Highlighted row is the winner.

Discount	Price after discount	Profit if won	Win chance	Expected profit	Worst case	Breaches floor?
0.0%	$20,911,979,419	-$6,201,889,505	36.9%	-$2,308,535,484	-$8,004,553,030	100% chance
2.5%	$20,389,179,933	-$6,706,571,208	38.3%	-$2,555,567,798	-$8,539,590,163	100% chance
5.0%	$19,866,380,448	-$7,245,794,854	40.1%	-$2,947,212,545	-$9,152,443,281	100% chance
7.5%	$19,343,580,962	-$7,777,393,386	43.1%	-$3,367,702,197	-$9,729,105,416	100% chance
10.0%	$18,820,781,477	-$8,296,980,644	47.5%	-$3,902,810,370	-$10,215,926,491	100% chance
12.5%	$18,297,981,991	-$8,764,796,546	53.2%	-$4,635,759,771	-$10,847,567,068	100% chance

Discount	Price after discount	Profit if won	Win chance	Expected profit	Worst case	Breaches floor?
15.0%	$17,775,182,506	-$9,326,860,393	60.3%	-$5,657,750,161	-$11,598,578,941	100% chance
17.5%	$17,252,383,021	-$9,818,238,699	67.8%	-$6,657,512,916	-$12,172,999,467	100% chance
20.0%	$16,729,583,535	-$10,361,989,236	75.2%	-$7,795,504,151	-$12,834,327,807	100% chance
22.5%	$16,206,784,050	-$10,897,730,890	81.3%	-$8,890,079,549	-$13,416,959,141	100% chance
25.0%	$15,683,984,564	-$11,429,738,228	85.9%	-$9,777,320,820	-$13,980,006,998	100% chance
27.5%	$15,161,185,079	-$11,963,274,230	89.2%	-$10,637,172,029	-$14,527,808,821	100% chance
30.0%	$14,638,385,593	-$12,510,726,465	91.4%	-$11,439,742,146	-$15,160,862,275	100% chance
32.5%	$14,115,586,108	-$13,003,729,323	92.7%	-$12,047,025,477	-$15,664,697,443	100% chance
35.0%	$13,592,786,622	-$13,544,412,280	93.4%	-$12,675,455,223	-$16,147,729,948	100% chance
37.5%	$13,069,987,137	-$14,015,285,361	93.9%	-$13,153,227,314	-$16,603,206,425	100% chance
40.0%	$12,547,187,651	-$14,564,680,547	94.1%	-$13,716,886,764	-$17,179,833,135	100% chance

Row 1 maximises expected profit.

The decision in one line

What changes if you approve the discount being asked for.

Column 1	Best (0%)	Asked (25%)	Difference
Discount	0.0%	25.0%	+25.0 pts
Win chance	36.9%	85.9%	+49.0 pts
Profit if won	-$6,201,889,505	-$11,429,738,228	-$5,227,848,723
Expected profit	-$2,308,535,484	-$9,777,320,820	-$7,468,785,336
Range where the answer is the same	0%	outside it	margin of error $126.13M

The range where the answer is the same

Every discount we tested, how far it falls short of the best, and whether that shortfall is real money or just the model's own margin of error.

Discount	Expected profit	Short of the best	Inside the range?	What it means
0.0%	-$2,308,535,484	$0	Inside	safe to approve
2.5%	-$2,555,567,798	$247,032,313	Outside	costs you $247.03M
5.0%	-$2,947,212,545	$638,677,060	Outside	costs you $638.68M
7.5%	-$3,367,702,197	$1,059,166,713	Outside	costs you $1,059.17M
10.0%	-$3,902,810,370	$1,594,274,886	Outside	costs you $1,594.27M
12.5%	-$4,635,759,771	$2,327,224,286	Outside	costs you $2,327.22M
15.0%	-$5,657,750,161	$3,349,214,676	Outside	costs you $3,349.21M
17.5%	-$6,657,512,916	$4,348,977,431	Outside	costs you $4,348.98M
20.0%	-$7,795,504,151	$5,486,968,667	Outside	costs you $5,486.97M
22.5%	-$8,890,079,549	$6,581,544,065	Outside	costs you $6,581.54M
25.0%	-$9,777,320,820	$7,468,785,336	Outside	costs you $7,468.79M
27.5%	-$10,637,172,029	$8,328,636,545	Outside	costs you $8,328.64M
30.0%	-$11,439,742,146	$9,131,206,662	Outside	costs you $9,131.21M
32.5%	-$12,047,025,477	$9,738,489,992	Outside	costs you $9,738.49M
35.0%	-$12,675,455,223	$10,366,919,738	Outside	costs you $10,366.92M
37.5%	-$13,153,227,314	$10,844,691,829	Outside	costs you $10,844.69M
40.0%	-$13,716,886,764	$11,408,351,279	Outside	costs you $11,408.35M

Highlighted rows are the range where the answer is the same (0%) - within about $126.13M of the best.

Recommended discount per deal (top 25)

The same logic applied to each deal individually.

Deal	Customer	Value	Margin	Best discount	Expected profit
Q-000011589	Acct-226235	$24,000,000	-38.2%	0%	-$3,391,345
Q-000009396	Acct-223884	$24,000,000	-39.4%	0%	-$3,490,019
Q-000009047	Acct-090640	$24,000,000	-28.1%	0%	-$2,491,751
Q-000008091	Acct-110274	$19,296,863	-39.7%	0%	-$2,829,654
Q-000007533	Acct-062342	$16,052,463	-14.0%	0%	-$832,133
Q-000018826	Acct-138872	$14,565,883	-9.4%	0%	-$505,239
Q-000002411	Acct-019732	$14,340,308	-17.7%	0%	-$938,218
Q-000006273	Acct-125458	$14,145,541	-11.5%	0%	-$599,155
Q-000005246	Acct-233590	$14,094,974	-15.4%	0%	-$802,001
Q-000013379	Acct-205409	$13,878,192	-33.0%	0%	-$1,693,461
Q-000004222	Acct-219531	$13,822,773	-10.9%	0%	-$555,899
Q-000009736	Acct-113136	$13,452,114	-26.1%	0%	-$1,295,886
Q-000017300	Acct-007220	$13,422,614	-17.5%	0%	-$868,789
Q-000000139	Acct-108061	$13,011,806	-41.2%	0%	-$1,979,664
Q-000009880	Acct-201403	$12,782,137	-23.9%	0%	-$1,127,369
Q-000001755	Acct-058430	$12,706,362	-9.2%	0%	-$430,973
Q-000001411	Acct-122168	$12,567,749	-33.8%	0%	-$1,569,967
Q-000010633	Acct-176419	$12,154,931	-10.9%	0%	-$488,694
Q-000013798	Acct-106373	$11,654,026	-31.0%	0%	-$1,333,799
Q-000010069	Acct-070297	$11,489,319	-9.7%	0%	-$411,261
Q-000013273	Acct-130561	$11,410,414	-31.5%	0%	-$1,327,590
Q-000007138	Acct-039861	$11,334,211	-9.3%	0%	-$390,313
Q-000005723	Acct-178920	$11,250,338	-42.0%	0%	-$1,746,544
Q-000001338	Acct-100033	$11,225,300	-22.2%	0%	-$921,328
Q-000009264	Acct-188657	$11,173,578	-32.8%	0%	-$1,353,554

What we recommend

No discount makes money on these numbers: even the least-bad one (0%) loses $2,308.54M. The deal is unprofitable before any discount is given - the fix is the cost to deliver or the price, not the size of the concession.

Risk High 100 out of 100 | Confidence: High

Why

• Only 0% stands out from the rest - the model cannot tell any neighbouring discount apart from it.

• Expected profit peaks at 0% (-$2,308.54M).

• The requested 25% delivers -$9,777.32M - $7,468.79M less than the peak, which is more than the model's margin of error.

• Win probability only improves +49 points for that giveaway.

What to do

1. Do not approve a discount on these numbers. At the best level we tested (0%) the deal still loses $2,308.54M.

2. Fix the deal first: the average margin in your file is -26.5%, so the cost to deliver has to come down (or the price go up) before any discount makes sense.

3. Above 0% (the top of the safe range), require VP sign-off: the 25% being asked for costs about $298.75M per point of discount.

4. At 25% there is a 100% chance of breaching the 12.0% margin floor - add a volume or term commitment to earn the deeper discount.

The trade-off: You are trading $7,468.79M of expected profit for +49 points of win probability.

What we assumed and what to watch

Worth knowing before you act

• Your history runs at a negative average margin (-26.5%): on average each deal costs more to deliver than it sells for. Every discount therefore loses money, and the result shows the least-bad option rather than a good one. Treat this as a problem with the deal's cost or price, not with the discount.

• At 25% discount there is a 100% chance the deal actually costs you money to deliver even if you win it.

Assumptions

• List value $20,911,979,419, cost $27,110,892,901 (-26.5% standard margin).

• Consumption effects: 0.2% of value is consumed as unmetered burst you still pay for; 4 months of rollover defers revenue.

• Win probability curve: from 35.0% at full price toward a ceiling of 95.0%, halfway at 16.6% discount (source: 20000 of your past deals).

• Win probability uncertainty: plus or minus 5 points.

• Cost volatility: plus or minus 6% of cost.

• Margin floor: 12.0%.

• Two discounts are treated as giving the same answer when the gap between their expected profits is smaller than twice the model's own margin of error at both discounts added together (about $126.13M at the best one), or smaller than 1% of the best profit - whichever is larger.

• The range where the answer is the same is 0%, taken from the 17 discount levels we tested.

• 10,000 simulations per discount level; seed 42 (re-running gives identical numbers).

• Deal / quote ID -> your 'deal_id' column

• Customer name -> your 'customer' column

• Deal value (list price) -> your 'deal_value' column

• Cost to deliver -> your 'cost' column

• Standard margin (%) -> your 'margin_pct' column

• Discount given (%) -> your 'discount_pct' column

• Won? (1/0) -> your 'won' column

• Hit a consumption tier accelerator -> your 'tier_accelerator_hit' column

• Unmetered burst used -> your 'burst_allowance_pct' column

• Rollover window -> your 'rollover_window_months' column

Appendix: the last steps of the run

Every line was produced while the model ran. Re-running with the same seed replays it exactly.

Time	Stage	What happened
0.00s	prepare	Hit a consumption tier accelerator -> your 'tier_accelerator_hit' column
0.00s	prepare	Unmetered burst used -> your 'burst_allowance_pct' column
0.00s	prepare	Rollover window -> your 'rollover_window_months' column
1.02s	prepare	Read 20,000 records into the model.

Time	Stage	What happened
1.02s	simulate	Running the simulation…
1.02s	read	Reading the deal you gave us...
1.04s	read	Found 20000 deal(s) worth $20,911.98M at an average -26.5% margin.
1.04s	read	Your deals run at a negative average margin (-26.5%): on average each deal costs more to deliver than it sells for, so every discount you test loses money. We will show you the least-bad option.
1.04s	prepare	Cost to deliver is about $26,454.77M.
1.06s	prepare	Added the consumption effects from your file: $656.12M of extra cost to serve.
1.06s	prepare	Working out how much a discount really moves the buyer...
1.49s	prepare	Learned the buyer's behaviour from 20000 past deals (they flip hardest around 16.6% discount).
1.49s	simulate	Testing 17 discount levels, 10,000 outcomes each (170,000 scenarios in total)...
1.49s	simulate	Simulated 11% of the discount levels...
1.49s	simulate	Simulated 23% of the discount levels...
1.49s	simulate	Simulated 35% of the discount levels...
1.49s	simulate	Simulated 47% of the discount levels...
1.49s	simulate	Simulated 58% of the discount levels...
1.49s	simulate	Simulated 70% of the discount levels...
1.50s	simulate	Simulated 82% of the discount levels...
1.50s	simulate	Simulated 94% of the discount levels...
1.50s	simulate	Simulated 100% of the discount levels...
1.50s	summarize	All discount levels tested. Looking for the sweet spot...
1.50s	summarize	Checked how much the answer would wobble if we ran it again - the range where the answer is the same is 0%.
1.52s	summarize	Done. Best discount is 0% (-$2,308.54M expected profit); the range where the answer is the same is 0%.